ethical hacking unit 5 12marks | MATLAB | Ask Gemini

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/ › MATLAB Drive

Editor

exp1.m × exp3.m ×

/MATLAB Drive/exp1.m

```
clc;
clear;
close all;

%% 1. Basic Parameters
fs = 1000;              % Sampling frequency in Hz
T = 1/fs;               % Sampling period
t = 0:T:1-T;            % Time vector

N = 4;                  % Butterworth filter order
fc_LP = 100;            % Low-pass cutoff frequency in Hz
fc_HP = 150;            % High-pass cutoff frequency in Hz

%% 2. Design Butterworth IIR Filters
% Normalized cutoff frequencies
Wn_LP = fc_LP/(fs/2);
Wn_HP = fc_HP/(fs/2);

% Low-pass Butterworth filter
[bLP, aLP] = butter(N, Wn_LP, 'low');

% High-pass Butterworth filter
```

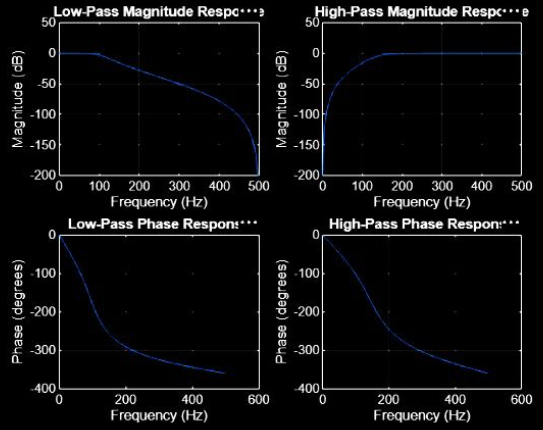
Command Window

```
Filter order              : 4
Cutoff frequency          : 150 Hz
Passband ripple           : 0.6855 dB
Stopband attenuation      : 9.3417 dB

Filter design and performance analysis completed.
>>
```

Figure 2: Magnitude and Phase Responses

FIGURE FORMAT TOOLS LAYOUT



Read "peppers.png" and display as grayscale

exp1.m

Ask Copilot

Validate generated output before use.

Editor: 100% UTF-8 LF Script Ln 3 Col 11